4.2 Human Blood Groups

Multiple allelism occurs in humans, particularly in the inheritance of different types of blood groups. The blood group inheritance in human can be understood by learning about antigens and antibodies. The composition of blood, different types of blood groups (ABO) the blood antigens and antibodies were discussed in chapter 7 of class XI.

4.2.1 ABO blood types

Multiple allele inheritance of ABO blood groups

Blood differs chemically from person to person. When two different incompatible blood types are mixed, agglutination (clumping together) of erythrocytes (RBC) occurs. The basis of these chemical differences is due to the presence of antigens (surface antigens) on the membrane of RBC and epithelial cells. Karl Landsteiner discovered two kinds of antigens called antigen 'A' and antigen 'B' on the surface of RBC's of human blood. Based on the presence or absence of these antigens three kinds of blood groups, type 'A', type 'B', and type 'O' (universal donor)were recognized. The fourth and the rarest blood group 'AB' (universal recipient) was discovered in 1902 by two of Landsteiner's students Von De Castelle and Sturli.

Bernstein in 1925 discovered that the inheritance of different blood groups in human beings is determined by a number of multiple allelic series. The three autosomal alleles located on chromosome 9 are concerned with the determination of blood group in any person. The gene controlling blood type has been labeled as 'L' (after the name of the discoverer, Landsteiner) or I (from isoagglutination). The I gene exists in three allelic forms, I^A, I^B and I^O. I^A specifies A antigen. I^B allele determines B antigen and I^O allele specifies no antigen. Individuals who possess these antigens in their fluids such as the saliva are called secretors.

Each allele (I^A and I^B) produces a transferase enzyme. I^A allele produces N-acetyl galactose transferase and can add N-acetyl galactosamine (NAG) and I^B allele encodes for the enzyme galactose transferase that adds galactose to the precursor (i.e., H substances) In the case of I^O/I^O allele no terminal transferase enzyme is produced and therefore called "null" allele and hence cannot add NAG or galactose to the precursor.

From the phenotypic combinations it is evident that the alleles I^A and I^B are dominant to I^O, but co-dominant to each other ($I^A=I^B$). Their dominance hierarchy can be given as ($I^A=I^B > I^O$). A child receives one of three alleles from each parent, giving rise to six possible genotypes and four possible blood types (phenotypes). The genotypes are I^AI^A, I^AI^O, I^BI^B, I^BI^O, I^AI^B and I^OI^O.



- Antigens similar to those found among human beings have been recognized in the blood of other organisms. A-type antigens have been found in chimpanzees and in gibbons, A, B and AB antigen in orangutans.
- New world monkeys (Platyrrhina) and lemurs have a substance similar but not identical with B antigen in humans.
- Three blood groups have been distinguished in cats with a genetic system similar to those in humans.
- The secretors (antigens found in the body fluids) can be detected in tears, saliva, urine, semen, gastric juice and in the milk of animals.

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